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Baton Rouge, LA 70817, USA

SUMMARY

Research-to-Product Bridge with 5+ years turning research into engineering impact, driving 0→1 innovation and scaling AI/ML solutions. I specialize in turning novel ideas in AI, XR, and cognitive science into deployable, scalable solutions by seamlessly connecting research, product, and engineering disciplines. Fluent in the languages of science (algorithms, study design), product (user needs, strategy), and engineering (APIs, pipelines, MLOps), I close the research-to-production gap and drive measurable impact. AWS and IBM AI Certified with proven experience leading cross-functional teams, guiding 0→1 innovation, and aligning usability, ethics, and scalability. I thrive in fast-paced, agile environments, collaborating across disciplines to accelerate innovation and consistently translate advanced research into products that scale.

SKILLS

- **Programming Languages:** Python, R, SQL.
- **IDEs:** Jupyter Notebook, VS Code, PyCharm, RStudio.
- **Research:** Experimental Design, Usability Human Computer Interaction Research, Survey Validation, Applied AI/ML Research, Multimodal Data Analysis, Time-Series Forecasting, A/B Testing, T-Testing, Educational & Workforce Analytics, Mixed-Methods Studies, 0-1 Product Research, Usability Testing, Ethnography, Iterative Design Validation, Developer Workflow Analysis, Journey Mapping, Concept Testing, Correlation Analysis, Thematic analysis.
- **AI/ML Frameworks:** Predictive modeling, Supervised ML (RF, XGBoost, SVM), Generative AI, learning analytics, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, CNN, LLM, BERT/GPT fine-tuning, Feature engineering Hyperparameter tuning, Bayesian Optimization, Keras, OpenAI APIs, Hugging Face Transformers.
- **Data Science & Engineering:** ETL design, Data Wrangling & Analysis, Data transformation, Statistical modeling and inference, Apache Spark, Hadoop, PySpark, Apache Kafka, Apache Airflow, BigQuery.
- **Databases:** PostgreSQL, MySQL, MongoDB.
- **Data Visualization:** Tableau, AWS QuickSight, Plotly, Matplotlib, Seaborn, Power BI.
- **Cloud Platforms & Deployment:** AWS (SageMaker, Glue, ECS, Fargate, CloudWatch, CodePipeline, BedRock), Azure ML, Azure, Kubernetes Service (AKS), GCP, Streamlit, CI/CD pipeline design, Docker, Kubernetes, Terraform, Watson Studio, AutoAI.
- **Project Management:** Git, GitHub, Agile Scrum, JIRA, Figma, UserTesting, Qualtrics, Optimizely, Miro, Lucidchart.

EXPERIENCE

- **Southern University and A & M College** Feb 2024 - Present
Postdoctoral - Research Manager (Applied Science & Data Science) Baton Rouge, LA
 - Translated learning science research into AI/VR training products, deploying 10+ immersive crisis leadership modules that trained 35+ leaders and improved decision-making speed by 25%.
 - Built, trained, and validated ML models including deep learning (CNN) and classical Machine Learning algorithms (SVM, XGBoost, Random Forest, KNN) on multimodal datasets (behavioral, survey, engagement), achieving 95.18% accuracy in classifying cognitive skill levels, directly informing adaptive curriculum design.
 - Conducted 0-to-1 product research for an AI-driven immersive training platform in the \$2M edtech sector, leading 45+ interviews, competitive analysis, and journey mapping to define MVP features.
 - Led concept testing of wireframes and prototypes, integrating user feedback to refine navigation and improve engagement by 25%.
 - Collaborated with AI engineers, software developers, and product managers to translate complex ML workflows into intuitive design flows, refining prototypes and boosting engagement by 25%.
 - Designed and executed mixed-methods UX studies (45+ interviews, surveys, A/B tests, usability evaluations), generating insights that drove a 36% adoption lift.
 - Mentored 10+ junior researchers in applied UX and AI methods, establishing research frameworks that increased team impact by 50%.
 - Developed and deployed a Generative AI safety assistant using RAG on campus-specific emergency response documents and weblinks, enabling real-time, research-backed guidance for 3,000+ students, faculty, and staff.
- **Amazon.com** June 2023 - Sept 2023
Data Scientist Intern, Mechatronics and Sustainable Package team Seattle, WA
 - Performed large-scale temporal and sensitivity analysis on 1M+ historical transactions and logistics using statistical and ML models, improving packaging risk detection and simulation robustness by 40%.
 - Developed a data transformation and enhancement framework using custom probability density functions and empirical distribution fitting, improving the simulation of packing time and robustness to cost variability by 40% and supporting scalable risk modeling.
 - Benchmarked machine learning and deep learning models to forecast storage capacity needs in automated sorting

- systems, enabling scenario-based planning that reduced costs by 15% while maintaining temporal variability.
- Collaborated with stakeholders across engineering, operations, and product teams to align risk modeling initiatives with business and sustainability objectives, uncovering key drivers through large-scale analysis, and providing clear, actionable insights.

• Southern University and A & M College

Aug 2018 - Dec 2023

Graduate Research Assistant – (Research science & Data Science)

Baton Rouge, LA

- Performed feature engineering and applied PCA to reduce 29 cognitive survey variables into 5 latent factors, uncovering actionable insights that guided curriculum design and optimized VR intervention strategies.
- Conducted UX research with 120 students, identifying 3 key engagement drivers that improved learning satisfaction by 25% and informed instructional design.
- Developed a predictive analytics app (Streamlit) with MLOps pipelines, achieving 92%
- Applied time-series forecasting (ARIMA) to education-to-workforce pipelines, producing policy-ready insights that informed scholarship allocation and recruitment strategies.
- Performed exploratory research (observations, interviews, surveys) with 400+ learners to uncover behaviors, cognitive patterns, and adoption barriers in AI/VR training.
- Conducted UX studies with 120 students (surveys, interviews), uncovering 3 engagement drivers that boosted satisfaction by 25% and enhanced AI immersive platform usability.
- Synthesized research findings into actionable insights, identifying correlations between AI/VR use and cognitive skill growth, which directly shaped product direction and informed design strategy.
- Defined MVP requirements for a predictive analytics app through 0→1 product research (workflow analysis, journey mapping, personas), aligning early prototypes with user and developer needs.

• IBM Data Science Academy

Dec 2021 - Dec 2021

Data Scientist

Remote

- Built ETL pipelines to transform insurance claims data, enabling accurate fraud detection.
- Developed and trained a Random Forest model using Scikit-learn to classify fraudulent claims.
- Deployed the real-time fraud detection model on IBM Watson Machine Learning for production use.

• Hacey health initiatives

Jan 2018 - Dec 2023

AI Consultant

Lagos, Nigeria

- Designed and implemented AI-driven models to optimize the planning and execution of public health programs, improving campaign targeting and resource allocation across underserved populations.
- Led the development of predictive analytics frameworks that resulted in a 25% increase in campaign efficiency and a 15% reduction in disease incidence within prioritized demographic segments.
- Built interactive, real-time dashboards using Tableau and Power BI to visualize health intervention metrics, enabling program managers to make data-informed decisions and improve response strategies by 30%.
- Collaborated with cross-functional teams – including epidemiologists, field officers, and policy advocates – to translate analytical insights into actionable public health interventions.
- Advised on the design and deployment of resilient internet infrastructure to support real-time data collection, remote health monitoring, and cloud-based AI operations in low-connectivity regions.
- Contributed to grant proposal development focused on leveraging AI for public health surveillance, disease prevention, and economic empowerment initiatives, resulting in competitive submissions to local and international funding bodies.

EDUCATION

• Southern University and A & M College, USA

Aug 2018 - Dec 2023

Ph.D. in Science and Mathematics Education (Major: Education Technology), Advisor: Dr. Albertha Lawson

USA

Ph.D. dissertation: Exploring the impact of immersive technology on engineering students' computational thinking skills

• University of Surrey, UK

Sep 2013 - Sept 2014

MSc in Mobile and Satellite Communication, Advisor: Dr. Fabien Heliot

UK

• Covenant University, Nigeria

Sep 2007 - July 2012

B.Eng in Computer Engineering

Nigeria

CERTIFICATIONS

• Amazon Web Service (AWS) Certified AI Practitioner

July 2025

Issued by Amazon Web Services (AWS) Training and Certification

[\[LINK\]](#)

• AWS Educate Machine Learning Foundations

June 2025

Issued by Amazon Web Services Training and Certification

[\[LINK\]](#)

• IBM Artificial Intelligence Practitioner

Jan 2022

Issued by IBM

[\[LINK\]](#)

• Enterprise Design Thinking Practitioner, Enterprise Design Thinking - Team Essentials for AI

Dec 2021

Issued by IBM

[\[LINK\]](#) [\[LINK\]](#)

• AWS Certified Cloud Practitioner

April 2021

Issued by Amazon Web Services Training and Certification

[\[LINK\]](#)

• Cisco Certified Network Associate Routing and Switching (CCNA Routing and Switching)

March 2011

Issued by Cisco

[\[LINK\]](#)